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Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Sequencing was performed on Illumina Novaseq 6000 S4. No custom software was used for data collection; standard Illumina sequencing protocols were followed.
Data analysis	We developed a custom computational pipeline for the processing of the TETRIS-seq sequencing data (as described in the manuscript). The pipeline used several custom Python scripts (Python 3.7.4) as well as the following publicly available software packages: Picard 2.18.15, Fulcrum Genomics Fgbio 1.3.0, Burrows-Wheeler Aligner (BWA) 0.7.17, the Genome Analysis Toolkit (GATK) 3.8, SAMtools 1.10 and 1.11, VarDictJava 1.8.2, Pindel 0.2.5b9, ANNOVAR (June 2020 release), pysam 0.23.3, pyfaidx 0.8.1.4, fuzzywuzzy 0.18.0, networkx 3.7 and Biopython 1.85. Reads were aligned to Homo_sapiens_assembly19 (broad b37/GRCh37). Mosaic chromosomal alteration calling additionally used ruptures 1.1.9 for change-point detection of segment boundaries. Pipeline code: https://github.com/the-blundell-lab/TETRIS-seq (archived on Zenodo: DOI: 10.5281/zenodo.22963603). Data analysis and figure generation used python 3.11.13 with NumPy 2.0.2, Pandas 2.3.0, Matplotlib 3.10.3, SciPy 1.16.0, scikit-learn 1.7.1, seaborn 0.13.2 and requests 2.32.3. Analysis code: https://github.com/the-blundell-lab/preAML_evolutionary_dynamics (archived on Zenodo: DOI: 10.5281/zenodo.23197961).

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

Somatic SNV, indel and FLT3-ITD calls are provided in Supplementary Table 6, somatic mCAs in Supplementary Table 7, and the corresponding somatic VCFs are deposited at Zenodo (DOI: 10.5281/zenodo.22262497). Participant-level clinical data, with ages and event timings given as completed units (rounded down) to protect participant confidentiality, are provided in Supplementary Table 10. The following are available under controlled access, deposited in the European Genome-phenome Archive (EGA) under study accession EGAS50000002168, comprising datasets EGAD50000003110 (SNV/indel panel) and EGAD50000003111 (mCA/chromosomal rearrangement panel), and released to approved researchers via a Data Access Committee administered by the corresponding authors: the raw sequencing reads from both the SNV/ indel and mCA/ chromosomal rearrangement panels; the germline variant calls from the SNV/ indel panel; the unfiltered per-sample variant calls at all positions covered by the SNV/ indel panel, and the corresponding annotated variant tables; the per-timepoint SNV calls output by the in silico noise-correction model and the annotated VarDictJava indel calls; the per-SNP genotype calls (SNP identity and B-allele frequency) and log-R-ratio data underlying the mCA calling; and the unrounded sample- and participant-level metadata required to reproduce the analyses. Data from UK Biobank (Fig. 5f) were obtained under Application 58387 and are available through the UK Biobank Access Management System. Cell-type proportions estimated by methylation-based deconvolution are provided in Supplementary Table 11; the underlying whole-genome bisulphite sequencing data are described in Fonseca et al.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	All participants are female. UKCTOCS enrolled post-menopausal women for ovarian cancer screening; the sex composition of the study is therefore determined by the parent trial design. Sex-disaggregated analysis is not applicable as there are no male participants.
Reporting on race, ethnicity, or other socially relevant groupings	Race and ethnicity were not collected as part of this study. UKCTOCS participants were recruited from the general UK population.
Population characteristics	47 post-menopausal women (aged 50-74 at enrolment) who were cancer-free at baseline and subsequently developed AML. 46 age- and timepoint-matched female controls who remained cancer-free throughout follow-up. All participants were enrolled in the UKCTOCS multimodal screening arm between 2001 and 2005.
Recruitment	Participants were identified from the UKCTOCS multimodal arm (202,638 women enrolled 2001-2005, of whom 50,640 were randomised to the multimodal arm involving annual blood sampling for up to 11 years). All women who were cancer-free at enrolment, subsequently developed AML during follow-up, and had serial blood samples available from at least 2 pre-AML timepoints were included (n = 47). Age- and timepoint-matched controls were selected from participants who remained cancer-free. One control was excluded due to subsequent MDS diagnosis.
Ethics oversight	UKCTOCS was reviewed and approved by the North West Multi-Centre Research Ethics Committee (MREC) (REC reference 00/8/34) and participants provided written informed consent for use of their samples and data in ethically approved secondary studies. This study, using UKCTOCS samples and data, was reviewed and approved by the South Central Berkshire B Research Ethics Committee (REC reference 18/SC/0481).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	47 pre-AML cases and 46 controls, with a median 5 blood samples per individual, range 2-10). Sample size was determined by the number of women developing AML in UKCTOCS multimodal arm who had sufficient serial samples available. No statistical method was used to predetermine sample size.
Data exclusions	One control was excluded after enrolment due to subsequent diagnosis of myelodysplastic syndrome (MDS). This exclusion criteria was pre-established: controls were required to remain free of haematological malignancy throughout the follow-up period.

Replication	This is an observational study using unique clinical samples; independent replication of the cohort is not possible. Technical validation of the TETRIS-seq pipeline was performed using synthetic dilution series and orthogonal validation methods as described in Supplementary Note 2, including assessment of sensitivity and specificity for SNVs, indels, and mosaic chromosomal alterations. Key evolutionary findings (somatic sweeps predicting AML risk) were independently validated in UK Biobank (n=335,090).
Randomization	This is an observational case-control study; participants were not randomized. Cases were identified by subsequent AML diagnosis and controls were matched by age and sample timepoints.
Blinding	The TETRIS-seq sequencing pipeline and variant calling were applied identically to all samples. Cases and controls were processed together on the same sequencing runs without knowledge of case/control status during sequencing and primary variant calling. The evolutionary inference (fitness and acquisition age estimation) was performed on individual cases using the variant calls as input.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☐	☒ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Clinical data

Policy information about clinical studies

All manuscripts should comply with the ICMJE guidelines for publication of clinical research and a completed CONSORT checklist must be included with all submissions.

Clinical trial registration	Samples were obtained from participants in UKCTOCS (ISRCTN22488978), a randomised controlled trial of ovarian cancer screening. The present study does not report UKCTOCS clinical trial outcomes; UKCTOCS provided the biobank and clinical follow-up data.
Study protocol	The UKCTOCS protocol is published (Menon et al., Lancet Oncol 2009; 10:327-340). The present study was approved by the South Central Berkshire B Research Ethics Committee (REC reference 18/SC/0481). The study protocol is described in the Methods section of the manuscript.
Data collection	Pre-diagnostic blood serum samples were collected annually between 2001 and 2011 as part of UKCTOCS. Clinical outcome data (AML diagnoses, other haematological and non-haematological malignancies, dates of death) were obtained through linkage to the National Cancer Registry, Hospital Episode Statistics (inHES), and death certificates.
Outcomes	The primary outcome was AML diagnosis, identified through ICD-10 coding (C92) via cancer registry linkage. AML diagnoses were confirmed across multiple independent data sources (National Cancer Registry, death certificates, and hospital episode statistics). This study does not report clinical trial outcomes from UKCTOCS.

Plants

Seed stocks	*Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.*
Novel plant genotypes	*Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.*
Authentication	*Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosiacism, off-target gene editing) were examined.*